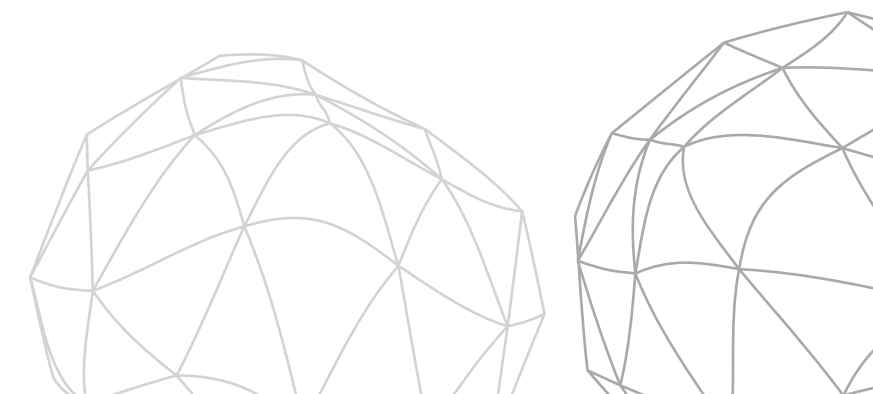




Grounded AI with SAP HANA Cloud Vector Engine





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github.com/nilpeksen/hana-vector-grounding



Agenda

1. The gap
2. From text to meaning - Vector Engine - embeddings, semantic search
3. From meaning to answer - RAG - retrieval, the gate, generation
4. The Demo
5. Direction



**The machine is running hotter than normal under full load,
have we seen something like this before?**

Not a threshold. Not an ID. A meaning.





SAP HANA Cloud Vector Engine



Value Search

WHERE temp > 90

Exact values.
Known attributes.



Meaning Search

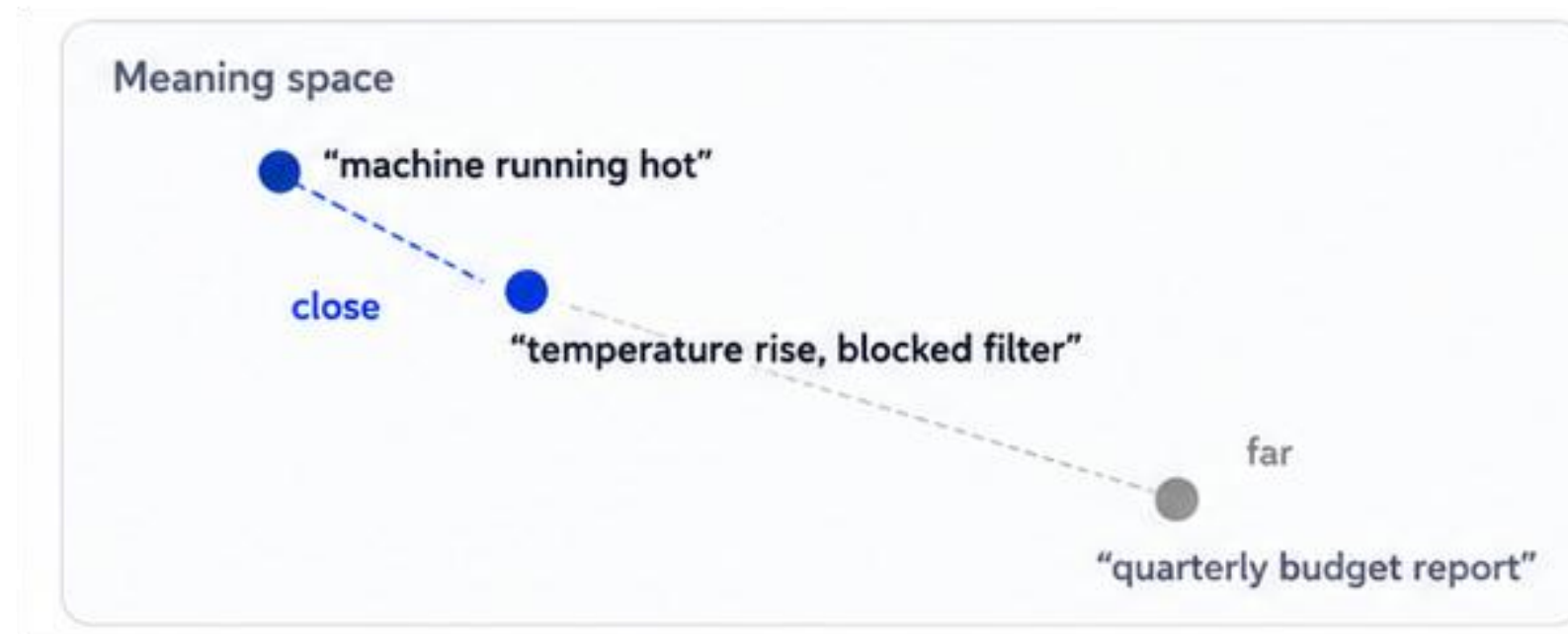
Most similar to “machine running hot”

Similarity by meaning
and works on any text.

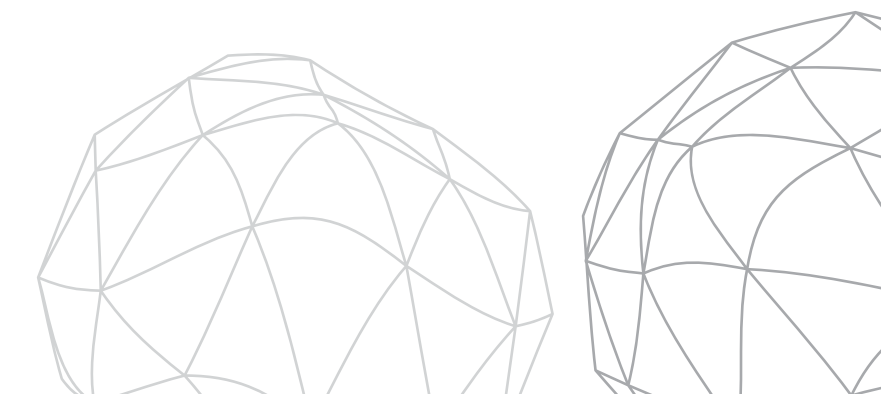




Vector Embeddings



Different words land close together. **Meaning, not spelling, sets the position.**





Retrieval Augmented Generation

Fine-tuning

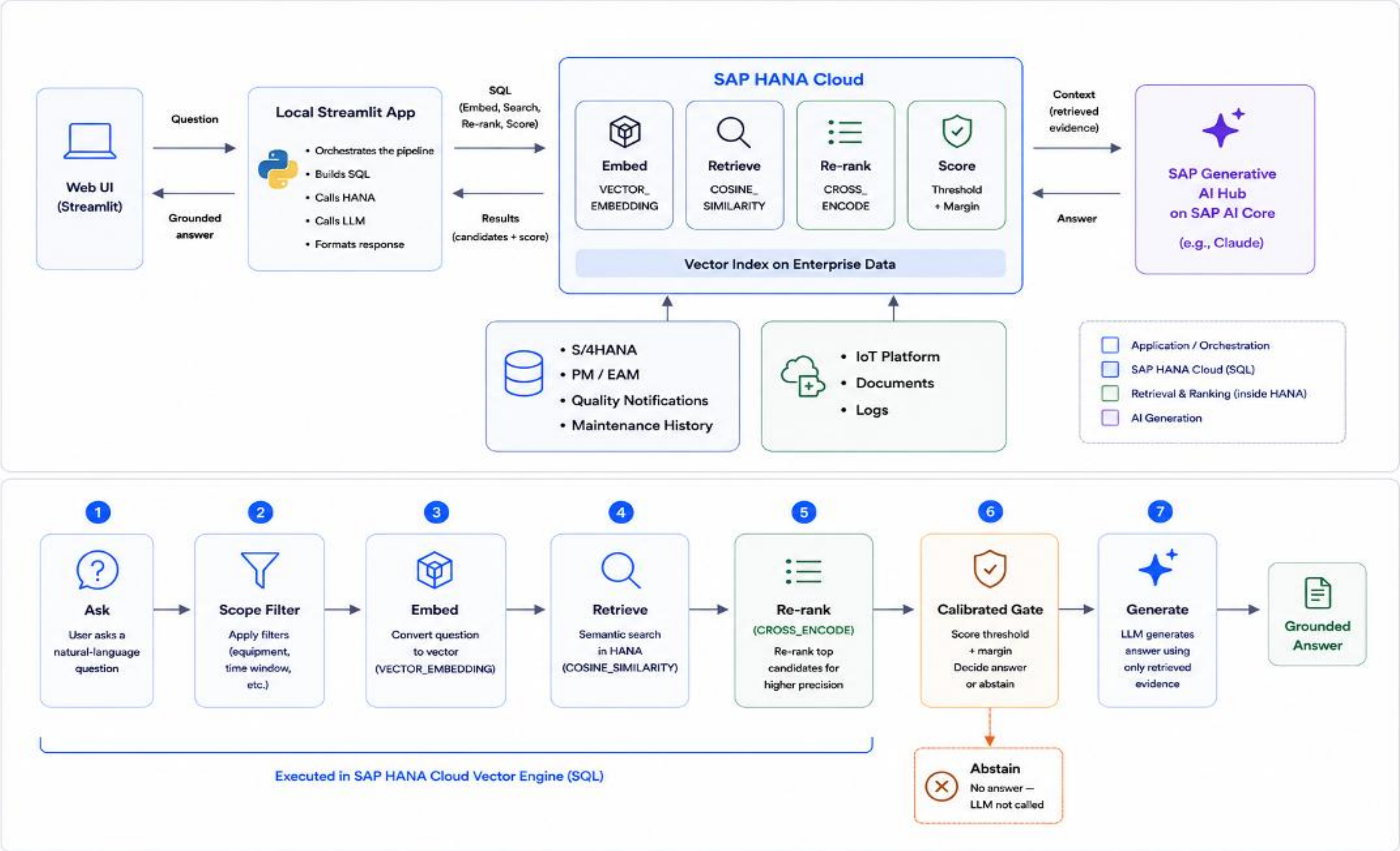
-  Retrain the model
-  Expensive, slow
-  Outdated after training

Retrieval

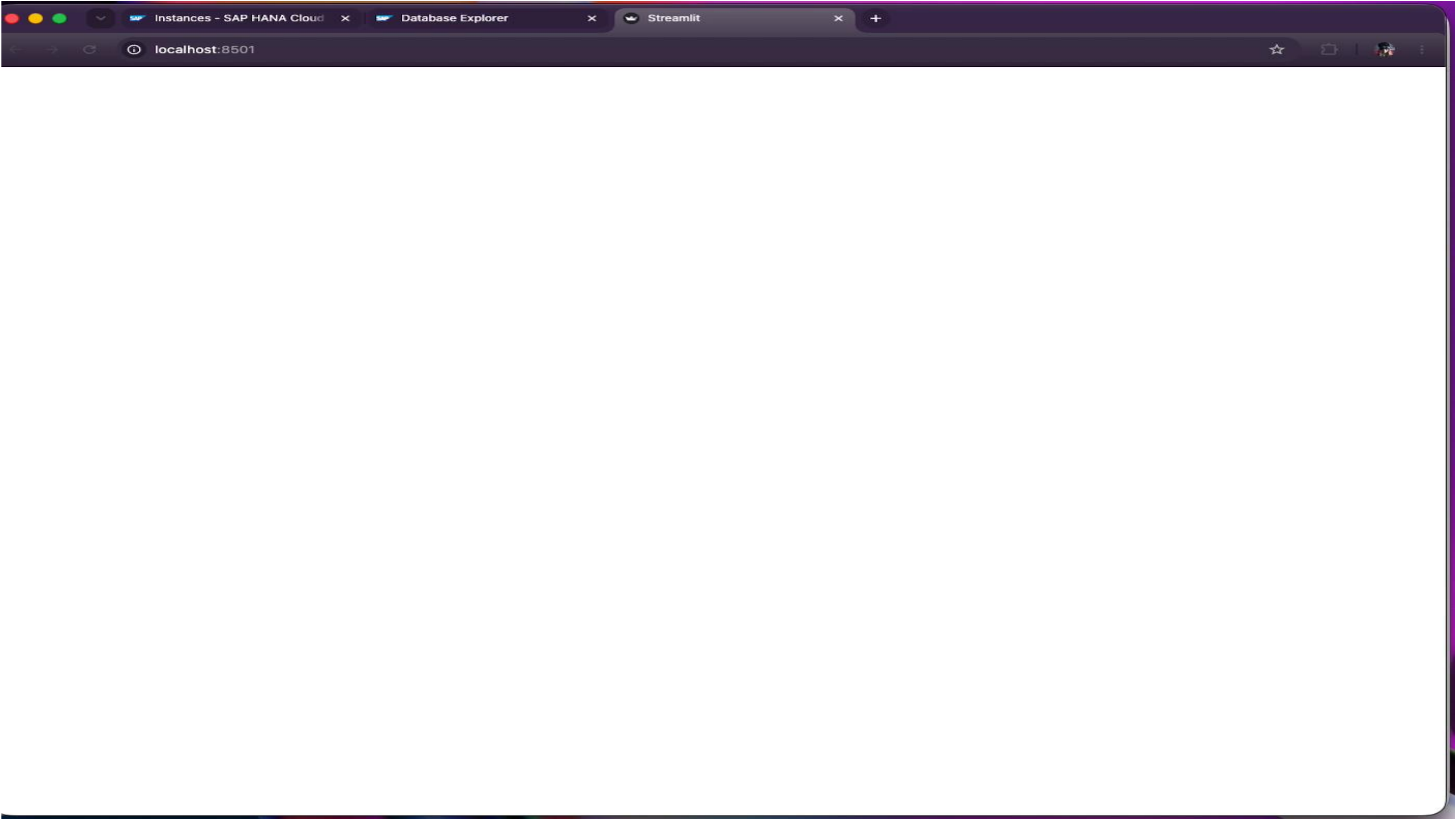
-  Keep the model
-  Ask, then retrieve
-  Stays in data



Demo Architecture & Pipeline



Demo



One Model, Three Outcomes



Question → LLM→ Generic Answer

From Data to Meaning

Grounded AI with SAP HANA Cloud Vector Engine

☐ Ground answers in SAP HANA Cloud

Question → Claude via SAP Generative AI Hub (training data only · no retrieval)

Maintenance engineer asks:

Compressor C-118 is running hotter than normal under full load. Have we seen something like this before, and what was the resolution?

Ask

WITHOUT GROUNDING — MODEL KNOWLEDGE ONLY

I don't have access to your specific maintenance history or equipment records for Compressor C-118. To help you effectively, I'd need you to check your:

Immediate troubleshooting steps for overheating compressor:

- Check cooling system (fans, heat exchangers, coolant levels)
- Verify oil levels and condition
- Inspect air intake filters
- Monitor discharge temperatures and pressures
- Check for proper ventilation around unit

To find historical data:

- CMMS/maintenance management system
- Equipment logbooks
- Previous work orders for C-118
- Similar compressor models in your facility

Common causes of compressor overheating:

- Dirty/clogged cooling systems
- Low oil levels or degraded oil
- Restricted air intake
- Excessive ambient temperature
- Worn internal components
- Improper loading/unloading cycles

Would you be able to share what specific temperatures you're seeing versus normal operating ranges? Also, has anything changed recently in the operating environment or maintenance schedule?

Question → HANA Vector Engine → LLM → Grounded Answer

From Data to Meaning

Grounded AI with SAP HANA Cloud Vector Engine

☒ Ground answers in SAP HANA Cloud

Question → scope filter (SQL) → VECTOR_EMBEDDING() → COSINE_SIMILARITY() → relevance threshold → Claude via SAP Generative AI Hub

> Retrieval scope — relational filter, applied before semantic ranking

Maintenance engineer asks:

Compressor C-118 is running hotter than normal under full load. Have we seen something like this before, and what was the resolution?

Ask

Retrieved from HANA — scoped by SQL, ranked by meaning

Notification	Equipment	Type	Date	Similarity	Long text
10004672	C-118	COMPRESSOR	2025-05-06	0.734	Blocked air intake filter on compressor C-118 restricted airflow and drove operating te
10034041	C-104	COMPRESSOR	2023-07-28	0.674	Compressor C-104 tripped on high discharge temperature after aftercooler fins were fr
10017374	P-207	PUMP	2024-11-19	0.656	Coolant circulation pump P-207 showed elevated casing temperature during extende
10007135	PMP-179	PUMP	2024-05-14	0.591	Pressure transmitter on pump PMP-179 drifted 0.4 bar against reference gauge. Transr
10008332	PMP-450	PUMP	2024-12-24	0.584	Gear pump PMP-450 cavitation noise reported by operators. Suction strainer cleaned

WITH GROUNDING — TOP 0.734, MARGIN 0.108 — PASSED CALIBRATED GATE — ANSWERS FROM OUR NOTIFICATION HISTORY

Yes, we have seen this exact issue before with compressor C-118.

According to Notification 10004672 dated 2025-05-06, compressor C-118 experienced overheating that drove operating temperature well above the acceptable range during sustained peak production output. The root cause was a blocked air intake filter that restricted airflow. The unit overheated and the thermal cutout tripped twice within one shift.

The resolution was:

- Filter element replaced
- Intake housing cleaned
- Inspection interval shortened

This matches your current symptom of C-118 running hotter than normal under full load, so I'd recommend checking the air intake filter first as it was the direct cause of the previous overheating issue.

Question → HANA Vector Engine → Gate → Abstain — LLM never called

From Data to Meaning

Grounded AI with SAP HANA Cloud Vector Engine

☒ Ground answers in SAP HANA Cloud

Question → scope filter (SQL) → VECTOR_EMBEDDING() → COSINE_SIMILARITY() → relevance threshold → Claude via SAP Generative AI Hub

> Retrieval scope — relational filter, applied before semantic ranking

Equipment types

COMPRESSOR × PUMP × MOTOR × FAN × HVAC × CONVEYOR × GEARBOX × MIXER × PRESS × PACKAGING ×

History window (years)

3

Maintenance engineer asks:

compressor in the plant is humming much louder than usual — but all readings look normal. Have we seen something like this before

Ask

Retrieved from HANA — scoped by SQL, ranked by meaning

Notification	Equipment	Type	Date	Similarity	Long text
10008332	PMP-450	PUMP	2024-12-24	0.625	Gear pump PMP-450 cavitation noise reported by operators. Suction strainer cleaned
10012290	PMP-546	PUMP	2025-07-14	0.585	Suction line vibration on dosing pump PMP-546. Pulsation dampener fitted.
10034041	C-104	COMPRESSOR	2023-07-28	0.583	Compressor C-104 tripped on high discharge temperature after aftercooler fins were fr
10017374	P-207	PUMP	2024-11-19	0.581	Coolant circulation pump P-207 showed elevated casing temperature during extende
10005234	PMP-235	PUMP	2026-02-13	0.575	Suction line vibration on dosing pump PMP-235. Pulsation dampener fitted.

GROUNDING CHECK FAILED — TOP 0.625, MARGIN 0.044 — BELOW CALIBRATED GATE

No sufficiently similar incident exists in the notification history for this question. Rather than generating an answer from weak context, the system abstains.

This is a feature, not a failure: a grounded system must know when it has nothing to ground on.

The Direction of Travel



More of the AI pipeline is moving toward the data - not away from it.





What to take home

1

Meaning is already a data type in HANA Cloud

REAL_VECTOR, VECTOR_EMBEDDING(), COSINE_SIMILARITY()
Native SQL, no ` new platform, no data leaving the database

2

Grounding is retrieval placed into the prompt

Same model, same question. The answer changed, the context changed.
No fine-tuning required.

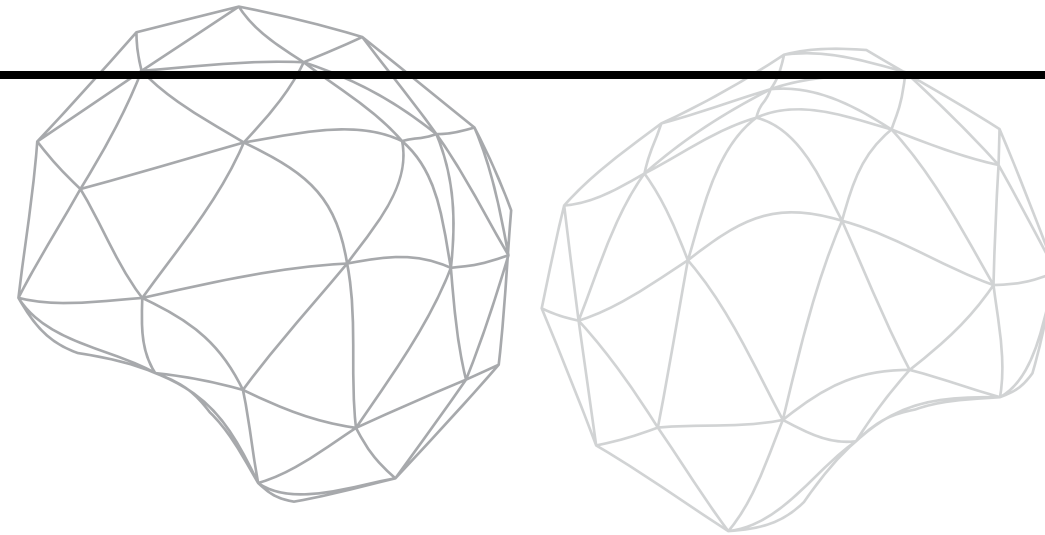
3

Your enterprise data is the moat, not the model

Outdated looking notification long texts nobody reads now searchable by meaning, and the difference between autocomplete and an answer you can act on.



From Data To Meaning



Thank You

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